

Search Theory and Unemployment

The Model

Partial Equilibrium

Infinite Horizon

Continuous time

Two states:

e = employed

u = unemployed (not working and searching for a job)

Preferences:

$$U = \mathbb{E}_0 \int_0^{\infty} e^{-\rho t} y(t) dt$$

where

$$y(t) = \begin{cases} w(t) & \text{if employed at } t \\ z & \text{if unemployed at } t \end{cases}$$

Discrete-time version: period of length h (small).

We'll let $h \rightarrow 0$ to get the continuous-time version

$\lambda h =$ prob. of receiving a job offer while unemployed

$1 - \lambda h =$ prob. of receiving no offer

$w \in [\underline{w}, \overline{w}] =$ wage attached to the job offer

$F(w) = \Pr\{W \leq w\}$ cdf wages ($f(w) \equiv F'(w)$ pdf)

$wh =$ wage income per period if employed at wage w

$\alpha h =$ prob. of being fired while employed

$1 - \alpha h =$ prob. keeping the job

The wage is constant as long as the job is kept

$zh = (b - c)h =$ net income per period if unemployed

Remark

More precisely, employment offers follow a time-homogeneous Poisson process with arrival rate λ . Hence, the probability of receiving n offers during time-interval h is:

$$q(n, h) = \frac{(\lambda h)^n}{n!} e^{-\lambda h}$$

Remark. $\mathbb{E}\{n\} = \lambda h$, $Var\{n\} = \lambda h$.

Hence:

$$q(0, h) = e^{-\lambda h}$$

$$q(1, h) = \lambda h e^{-\lambda h}$$

$$q(n \geq 2, h) = 1 - (1 + \lambda h) e^{-\lambda h}$$

A first-order Taylor approximation of $q(0, h) = e^{-\lambda h}$ around $h = 0$ gives:

$$q(0, h) \cong q(0, 0) + q_h(0, 0)(h - 0)$$

$$q(0, h) \cong e^{-\lambda 0} - \lambda e^{-\lambda 0}(h - 0) \Rightarrow$$

Then:

$$q(0, h) \cong 1 - \lambda h$$

A first-order Taylor approximation of $q(1, h) = \lambda h e^{-\lambda h}$ around $h = 0$ gives:

$$q(1, h) \cong q(1, 0) + q_h(1, 0)(h - 0)$$

$$q(1, h) \cong \lambda 0 e^{-\lambda 0} + (\lambda e^{-\lambda 0} - \lambda 0 e^{-\lambda 0} \lambda)(h - 0) \Rightarrow$$

$$q(1, h) \cong \lambda h$$

It follows that $q(n \geq 2, h) = 1 - q(0, h) - q(1, h) \cong 1 - (1 - \lambda h) - \lambda h \Rightarrow$

$$q(n \geq 2, h) \cong 0$$

Hence, during a sufficiently short period of time, an unemployed agent essentially receives either zero offers or just one offer.

During time-interval h , an employed worker is fired with probability $p(h) = 1 - e^{-\alpha h}$, and remains employed with probability $1 - p(h) = e^{-\alpha h}$.

A first-order Taylor approximation of $p(h)$ around $h = 0$ gives:

$$p(h) \cong p(0) + p'(0)(h - 0)$$

$$p(h) \cong 1 - e^{-\alpha 0} + \alpha e^{-\alpha 0}(h - 0)$$

$$p(h) \cong 1 - 1 + \alpha h \Rightarrow$$

$$p(h) \cong \alpha h$$

Then:

$$1 - p(h) \cong 1 - \alpha h$$

Discount Factor:

$$e^{-\rho h} = \frac{1}{e^{\rho h}}$$

A first-order Taylor approximation of $e^{\rho h}$ around $h = 0$ gives:

$$e^{\rho h} \cong e^{\rho 0} + e^{\rho 0} \rho (h - 0)$$

$$e^{\rho h} \cong 1 + \rho h$$

Then:

$$e^{-\rho h} \cong \frac{1}{1 + \rho h}$$

Back to the Model

Notice that the only decision to be made by an agent is to decide whether to accept an offer while unemployed (and stop looking for a job) or reject it (and keep looking for another opportunity). The agent is not allowed to search for a new job while employed (on-the-job search is ruled out by assumption).

We'll show that the optimal decision rule is to set a reservation wage and accept any offer with an attached wage higher or equal than the reservation wage.

Define:

$V_e(w, t, h)$ = value of employment at time t , with wage w , for a period of length h

$V_u(t, h)$ = value of unemployment at time t , for a period of length h .

Then:

$$V_e(w, t, h) = wh + \frac{1}{1 + \rho h} [\alpha h V_u(t + h, h) + (1 - \alpha h) V_e(w, t + h, h)]$$

Since the problem is stationary, $V_e(w, t, h)$ and $V_u(t, h)$ are the same for all t . Therefore, we can simply write $V_e(w, h)$ and $V_u(h)$. Then:

$$V_e(w, h) = wh + \frac{1}{1 + \rho h} [\alpha h V_u(h) + (1 - \alpha h) V_e(w, h)]$$

The expression above gives:

$$V_e(w, h) (1 - \frac{1 - \alpha h}{1 + \rho h}) = wh + \frac{\alpha h}{1 + \rho h} V_u(h)$$

$$V_e(w, h) \frac{(\rho + \alpha)h}{1 + \rho h} = wh + \frac{\alpha h}{1 + \rho h} V_u(h) \Rightarrow$$

$$V_e(w, h) = \frac{1 + \rho h}{\alpha + \rho} w + \frac{\alpha}{\alpha + \rho} V_u(h)$$

Continuous time: $h \rightarrow 0$

$$V_e(w) = \frac{w}{\alpha + \rho} + \frac{\alpha}{\alpha + \rho} V_u$$

where $V_e(w) \equiv V_e(w, 0)$ and $V_u \equiv V_u(0)$.

The expression above can be rewritten as follows:

$$\rho V_e(w) = w + \alpha[V_u - V_e(w)]$$

Asset-Pricing Interpretation

ρ = risk-free rate

$V_e(w)$ = current price

w = dividend

$\alpha[V_u - V_e(w)]$ = expected capital gain/loss

Risk-neutral investor $\Rightarrow \rho = \frac{w + \alpha(V_u - V_e)}{V_e}$

Or: $\rho = \frac{w + [\alpha V_u + (1 - \alpha)V_e] - V_e}{V_e}$

$V_u(h)$ must satisfy:

$$V_u(h) = zh + \frac{1}{1+\rho h} [(1-\lambda h)V_u(h) + \lambda h \mathbb{E} \max\{V_e(W, h), V_u(h)\}]$$

Then:

$$V_u(h)(1 - \frac{1-\lambda h}{1+\rho h}) = zh + \frac{\lambda h}{1+\rho h} \mathbb{E} \max\{V_e(W, h), V_u(h)\}$$

$$V_u(h) \frac{(\rho+\lambda)h}{1+\rho h} = zh + \frac{\lambda h}{1+\rho h} \mathbb{E} \max\{V_e(W, h), V_u(h)\}$$

$$V_u(h) = \frac{1+\rho h}{(\rho+\lambda)h} zh + \frac{1+\rho h}{(\rho+\lambda)h} \frac{\lambda h}{1+\rho h} \mathbb{E} \max\{V_e(W, h), V_u(h)\} \Rightarrow$$

$$V_u(h) = \frac{1+\rho h}{\rho+\lambda} z + \frac{\lambda}{\rho+\lambda} \mathbb{E} \max\{V_e(W, h), V_u(h)\}$$

Continuous time: $h \rightarrow 0$

$$V_u = \frac{z}{\rho + \lambda} + \frac{\lambda}{\rho + \lambda} \mathbb{E} \max\{V_e(W), V_u\}$$

From the expression above we get:

$$(\rho + \lambda)V_u = z + \lambda \mathbb{E} \max\{V_e(W), V_u\}$$

$$\rho V_u = z + \lambda \mathbb{E} \max\{V_e(W), V_u\} - \lambda V_u$$

$$\rho V_u = z + \lambda [\mathbb{E} \max\{V_e(W), V_u\} - V_u]$$

$$\rho V_u = z + \lambda \mathbb{E} \{\max\{V_e(W), V_u\} - V_u\} \Rightarrow$$

$$\rho V_u = z + \lambda \mathbb{E} \max\{V_e(W) - V_u, 0\}$$

Or:

$$\rho V_u = z + \lambda \mathbb{E} \max\{V_e(W), V_u\} - \lambda V_u$$

$$\rho = \frac{z + \lambda \mathbb{E} \max\{V_e(W), V_u\} - \lambda V_u}{V_u}$$

$$\rho = \frac{z + \lambda \mathbb{E} \max\{V_e(W), V_u\} - \lambda V_u + V_u - V_u}{V_u} \Rightarrow$$

$$\rho = \frac{z + [\lambda \mathbb{E} \max\{V_e(W), V_u\} + (1 - \lambda)V_u] - V_u}{V_u}$$

Remark. Analogous asset-pricing interpretation.

Define:

$$\omega \equiv \rho V_u$$

From $V_e(w) = \frac{w}{\alpha+\rho} + \frac{\alpha}{\alpha+\rho}V_u$ we get:

$$V_e(w) - V_u = \frac{w}{\alpha+\rho} + \frac{\alpha}{\alpha+\rho}V_u - V_u$$

$$V_e(w) - V_u = \frac{w}{\alpha+\rho} - \frac{\rho V_u}{\alpha+\rho}$$

$$V_e(w) - V_u = \frac{w}{\alpha+\rho} - \frac{\omega}{\alpha+\rho} \Rightarrow$$

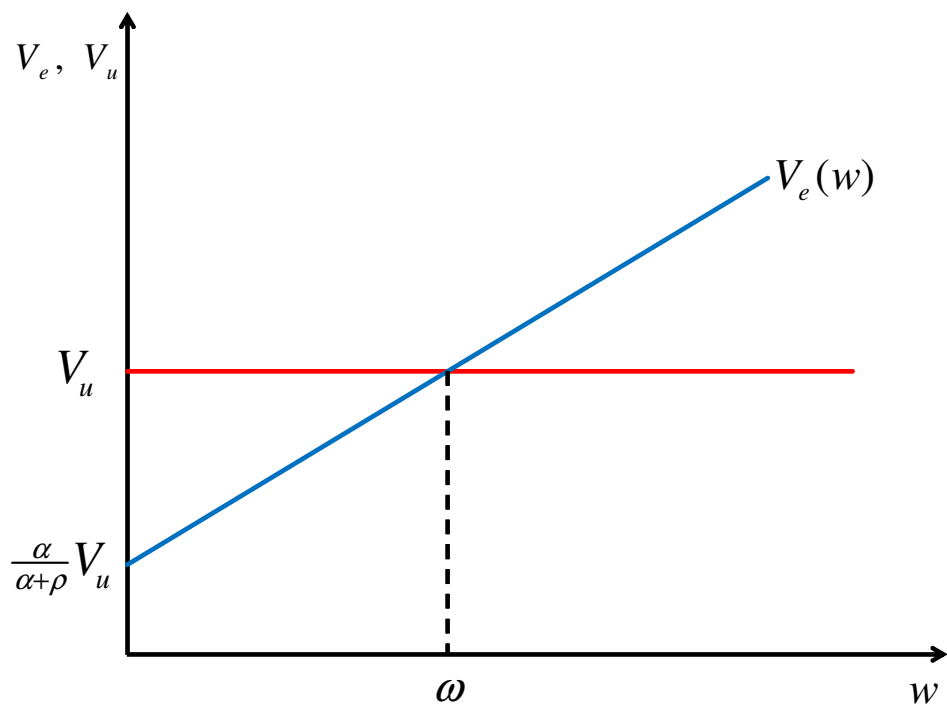
$$V_e(w) - V_u = \frac{w - \omega}{\rho + \alpha}$$

Therefore:

$$V_e(w) > V_u \text{ if and only if } w > \omega$$

Hence, it is optimal to accept an employment offer and become employed (stop searching) when the wage attached to the offer is above ω . That's why ω is called the *reservation wage*.

Plot



Using the definition of ω we can rewrite $\rho V_u = z + \lambda \mathbb{E} \max\{V_e(W) - V_u, 0\}$ as follows:

$$\omega - z = \lambda \mathbb{E} \max\{V_e(W) - V_u, 0\}$$

Using $V_e(w) - V_u = \frac{w-\omega}{\rho+\alpha}$ we get:

$$\omega - z = \lambda \mathbb{E} \max\left\{\frac{W-\omega}{\rho+\alpha}, 0\right\} \Rightarrow$$

$$\omega - z = \frac{\lambda}{\rho + \alpha} \mathbb{E} \max\{W - \omega, 0\}$$

The expression above determines the value of ω (and then the value of V_u , since $V_u = \frac{\omega}{\rho}$)

Interpretation:

l-h-s: marginal cost of search when wage offer is ω

r-h-s: marginal return to search when wage offer is ω

Notice that:

$$\mathbb{E} \max\{W - \omega, 0\} = \int_{\underline{w}}^{\bar{w}} \max\{w - \omega, 0\} f(w) dw$$

$$\mathbb{E} \max\{W - \omega, 0\} = \int_{\underline{w}}^{\omega} 0 f(w) dw + \int_{\omega}^{\bar{w}} (w - \omega) f(w) dw$$

$$\mathbb{E} \max\{W - \omega, 0\} = \int_{\omega}^{\bar{w}} (w - \omega) f(w) dw$$

Then:

$$\omega - z = \frac{\lambda}{\rho + \alpha} \int_{\omega}^{\bar{w}} (w - \omega) f(w) dw$$

Or:

$$\omega - z = \frac{\lambda[1 - F(\omega)]}{\rho + \alpha} \int_{\omega}^{\bar{w}} (w - \omega) \frac{f(w)}{1 - F(\omega)} dw$$

$$\omega - z = \frac{\lambda[1 - F(\omega)]}{\rho + \alpha} \left[\int_{\omega}^{\bar{w}} w \frac{f(w)}{1 - F(\omega)} dw - \omega \int_{\omega}^{\bar{w}} \frac{f(w)}{1 - F(\omega)} dw \right] \Rightarrow$$

$$\omega - z = \frac{\lambda[1 - F(\omega)]}{\rho + \alpha} [\mathbb{E}\{w \mid w \geq \omega\} - \omega]$$

Integrating $\int_{\omega}^{\bar{w}} (w - \omega) f(w) dw$ by parts we get:

$$u = w - \omega, \quad v = F(w)$$

$$du = dw, \quad dv = F'(w) dw = f(w) dw$$

$$\int_{\omega}^{\bar{w}} (w - \omega) f(w) dw = (w - \omega) F(w) \Big|_{\omega}^{\bar{w}} - \int_{\omega}^{\bar{w}} F(w) dw$$

$$\int_{\omega}^{\bar{w}} (w - \omega) f(w) dw = (\bar{w} - \omega) F(\bar{w}) - (\omega - \omega) F(\omega) - \int_{\omega}^{\bar{w}} F(w) dw$$

$$\int_{\omega}^{\bar{w}} (w - \omega) f(w) dw = (\bar{w} - \omega) - \int_{\omega}^{\bar{w}} F(w) dw$$

$$\int_{\omega}^{\bar{w}} (w - \omega) f(w) dw = \int_{\omega}^{\bar{w}} dw - \int_{\omega}^{\bar{w}} F(w) dw$$

$$\int_{\omega}^{\bar{w}} (w - \omega) f(w) dw = \int_{\omega}^{\bar{w}} [1 - F(w)] dw$$

Then:

$$\omega - z = \frac{\lambda}{\rho + \alpha} \int_{\omega}^{\bar{w}} [1 - F(w)] dw$$

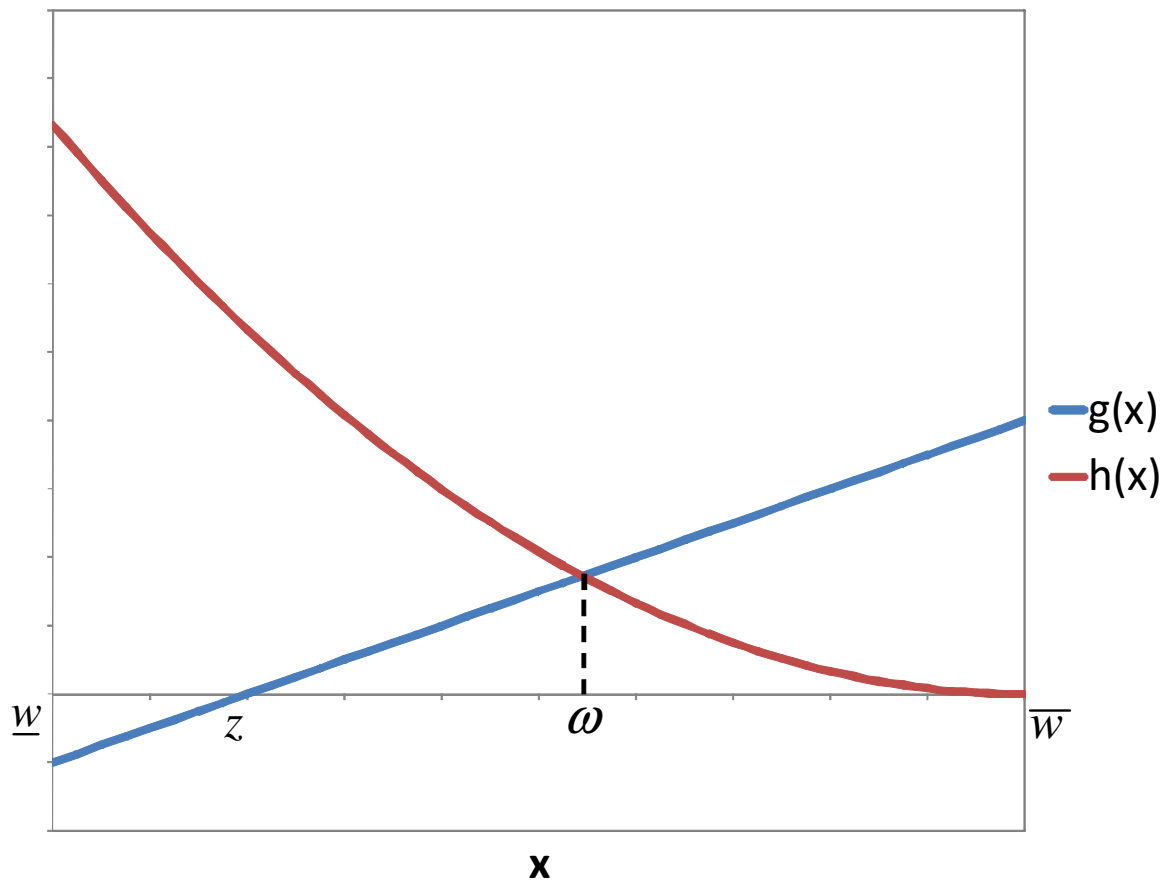
Define:

$$g(x) = x - z \quad \forall x \in [\underline{w}, \bar{w}]$$

$$h(x) = \frac{\lambda}{\rho + \alpha} \int_x^{\bar{w}} [1 - F(w)] dw \quad \forall x \in [\underline{w}, \bar{w}]$$

$$= \frac{\lambda}{\rho + \alpha} \int_x^{\bar{w}} (w - x) f(w) dw \quad \forall x \in [\underline{w}, \bar{w}]$$

Then:



Remark:

$$h(\underline{w}) = \frac{\lambda}{\alpha + \rho}(\mu - \underline{w}) > 0 \quad \text{where } \mu \equiv \mathbb{E}W$$

$$h(\overline{w}) = 0$$

$$h'(x) = \frac{-\lambda}{\alpha + \rho}[1 - F(x)] \quad \begin{cases} < 0 & \forall x \in [\underline{w}, \overline{w}) \\ = 0 & \text{at } x = \overline{w} \end{cases}$$

$$h''(x) = \frac{\lambda}{\alpha + \rho}f(x) > 0 \quad \forall x \in [\underline{w}, \overline{w}]$$

Unemployment

Define:

$$\theta_u \equiv \lambda[1 - F(\omega)]$$

$$\theta_e \equiv \alpha$$

Then:

$$\theta_u h = \lambda h [1 - F(\omega)]$$

$$\theta_e h = \alpha h$$

We have:

$$\lambda h = \text{prob. offer}$$

$$1 - F(\omega) = \text{prob. accepting an offer } (w > \omega) \\ \text{conditional on receiving one}$$

$$\theta_u h = \text{prob. leaving unemployment}$$

$$\theta_e h = \alpha h = \text{prob. leaving employment}$$

Consider an economy populated by a continuum of mass L of *ex ante* identical agents.

Define:

$$\begin{aligned} E(t) &= \text{employment at } t \\ U(t) &= \text{unemployment at } t \\ L &= \text{labor force (fixed)} \end{aligned}$$

Then:

$$E(t) + U(t) = L \text{ for all } t$$

Then:

$$U(t + h) - U(t) = \theta_e h E(t) - \theta_u h U(t)$$

Dividing by h :

$$\frac{U(t + h) - U(t)}{h} = \theta_e E(t) - \theta_u U(t)$$

Letting $h \rightarrow 0$:

$$\dot{U}(t) = \theta_e E(t) - \theta_u U(t)$$

In steady-state: $E(t) = E$, $U(t) = U$, and $\dot{U}(t) = 0$.

Define:

$$u \equiv \frac{U}{L} = \text{steady-state unemployment rate}$$

Setting $\dot{U}(t) = 0$ and using $E = L - U$, we obtain:

$$0 = \theta_e E - \theta_u U$$

$$0 = \theta_e (L - U) - \theta_u U$$

$$0 = \theta_e L - (\theta_e + \theta_u)U$$

$$(\theta_e + \theta_u)U = \theta_e L$$

$$\frac{U}{L} = \frac{\theta_e}{\theta_e + \theta_u} \Rightarrow$$

$$\begin{aligned} u &= \frac{\theta_e}{\theta_e + \theta_u} \\ &= \frac{\alpha}{\alpha + \lambda[1 - F(\omega)]} \end{aligned}$$

Expected duration of employment and unemployment spells

Let T_u denote the (random) duration of a completed unemployment spell. Then (see the Appendix):

$$\begin{aligned} D_u(t) &\equiv \Pr\{T_u \leq t\} \\ &= 1 - e^{-\theta_u t} \end{aligned}$$

The associated density function, $d_u(t) = D'_u(t)$, is:

$$d_u(t) = \theta_u e^{-\theta_u t}$$

From the expression above we can calculate the expected duration of completed unemployment spells. We get:

$$\begin{aligned}\mathbb{E}T_u &= \int_0^{\infty} t\theta_u e^{-\theta_u t} dt \\ &= \frac{1}{\theta_u}\end{aligned}$$

Remark. Integrate by parts using $u = t$ and $v = -e^{-\theta_u t}$.

Analogously, denoting by T_e the duration of completed employment spells, we get:

$$\mathbb{E}T_e = \frac{1}{\theta_e}$$

Substituting $\mathbb{E}T_u = \frac{1}{\theta_u}$ and $\mathbb{E}T_e = \frac{1}{\theta_e}$ into $u = \frac{\theta_e}{\theta_e + \theta_u}$ we get:

$$\begin{aligned}u &= \frac{\frac{1}{\mathbb{E}T_e}}{\frac{1}{\mathbb{E}T_e} + \frac{1}{\mathbb{E}T_u}} = \frac{\frac{1}{\mathbb{E}T_e}}{\frac{\mathbb{E}T_u + \mathbb{E}T_e}{\mathbb{E}T_e \mathbb{E}T_u}} \Rightarrow \\ u &= \frac{\mathbb{E}T_u}{\mathbb{E}T_u + \mathbb{E}T_e}\end{aligned}$$

Comparative Statics

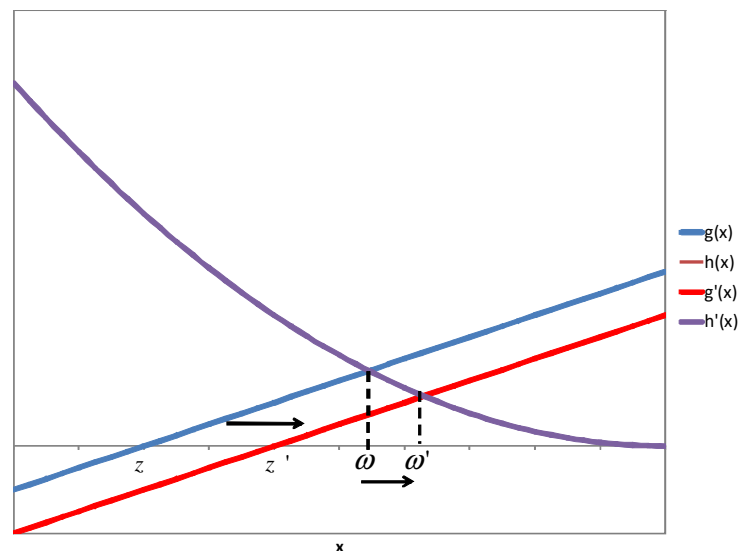
Recall that:

$$\omega - z = \frac{\lambda}{\rho + \alpha} \int_{\omega}^{\bar{w}} [1 - F(w)] dw$$

Change in z

$$\frac{\partial \omega}{\partial z} = \frac{\rho + \alpha}{\rho + \alpha + \theta_u} \in (0, 1)$$

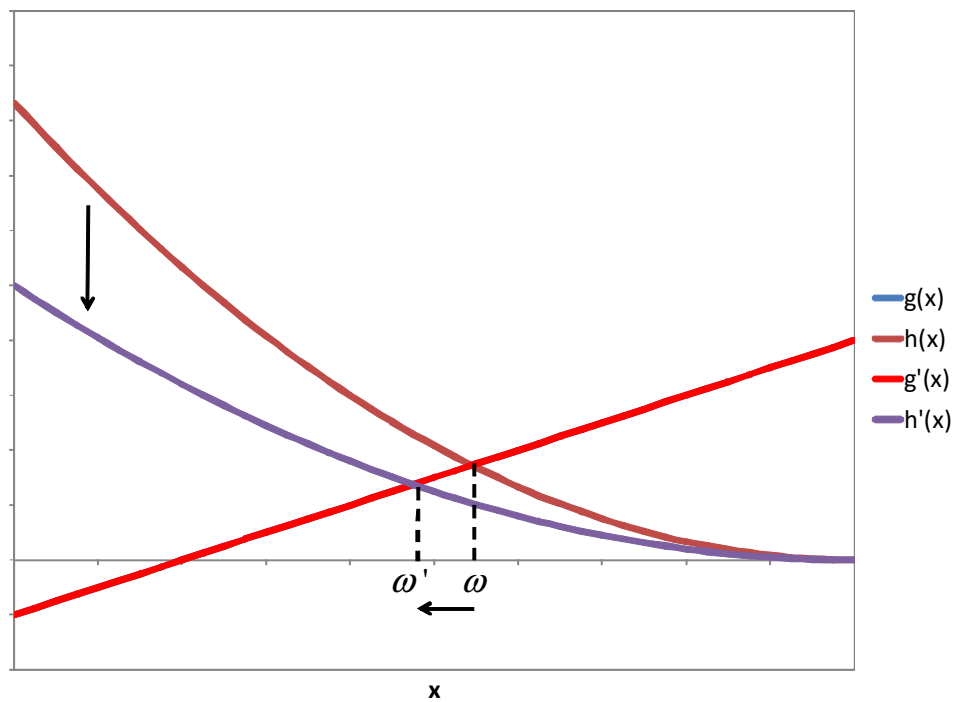
$$\begin{aligned} \uparrow z = b - c &\Rightarrow \uparrow \omega \Rightarrow \uparrow F(\omega) \Rightarrow \downarrow 1 - F(\omega) \Rightarrow \\ \downarrow \theta_u = \lambda[1 - F(\omega)] &\Rightarrow \uparrow u = \frac{\theta_e}{\theta_e + \theta_u} \end{aligned}$$



Change in ρ

$$\frac{\partial \omega}{\partial \rho} = -\frac{\omega - z}{\rho + \alpha + \theta_u} < 0$$

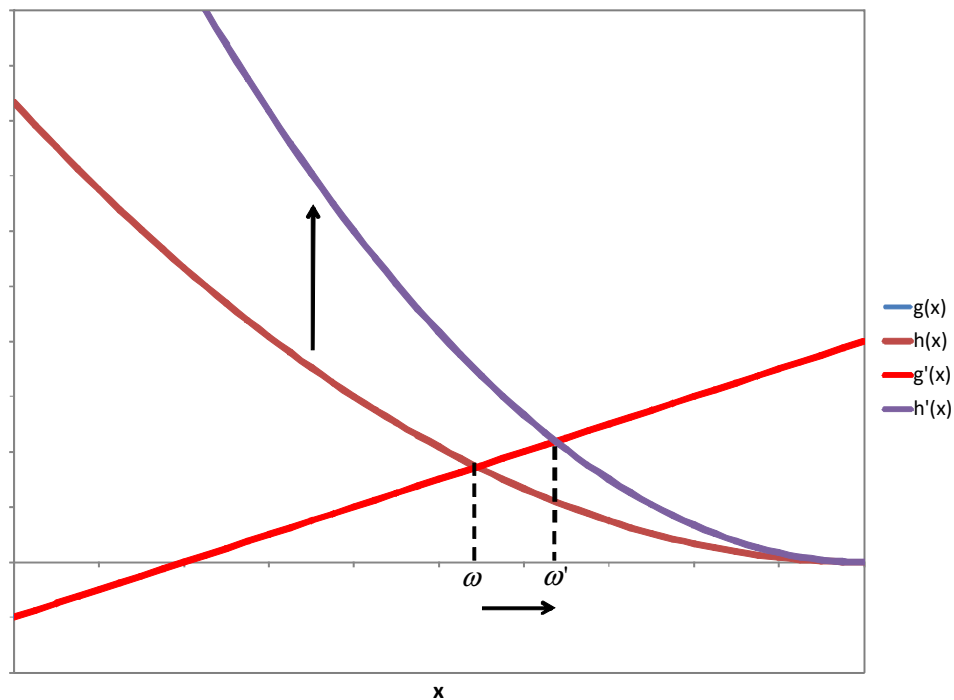
$$\uparrow \rho \Rightarrow \downarrow \omega \Rightarrow \downarrow F(\omega) \Rightarrow \uparrow 1 - F(\omega) \Rightarrow \uparrow \theta_u = \lambda[1 - F(\omega)] \Rightarrow \downarrow u = \frac{\theta_e}{\theta_e + \theta_u}$$



Change in λ

$$\begin{aligned}\frac{\partial \omega}{\partial \lambda} &= \frac{1}{\rho + \alpha + \theta_u} \int_{\omega}^{\bar{w}} [1 - F(w)] dw > 0 \\ &= \frac{\rho + \alpha}{\rho + \alpha + \theta_u} \frac{\omega - z}{\lambda} > 0\end{aligned}$$

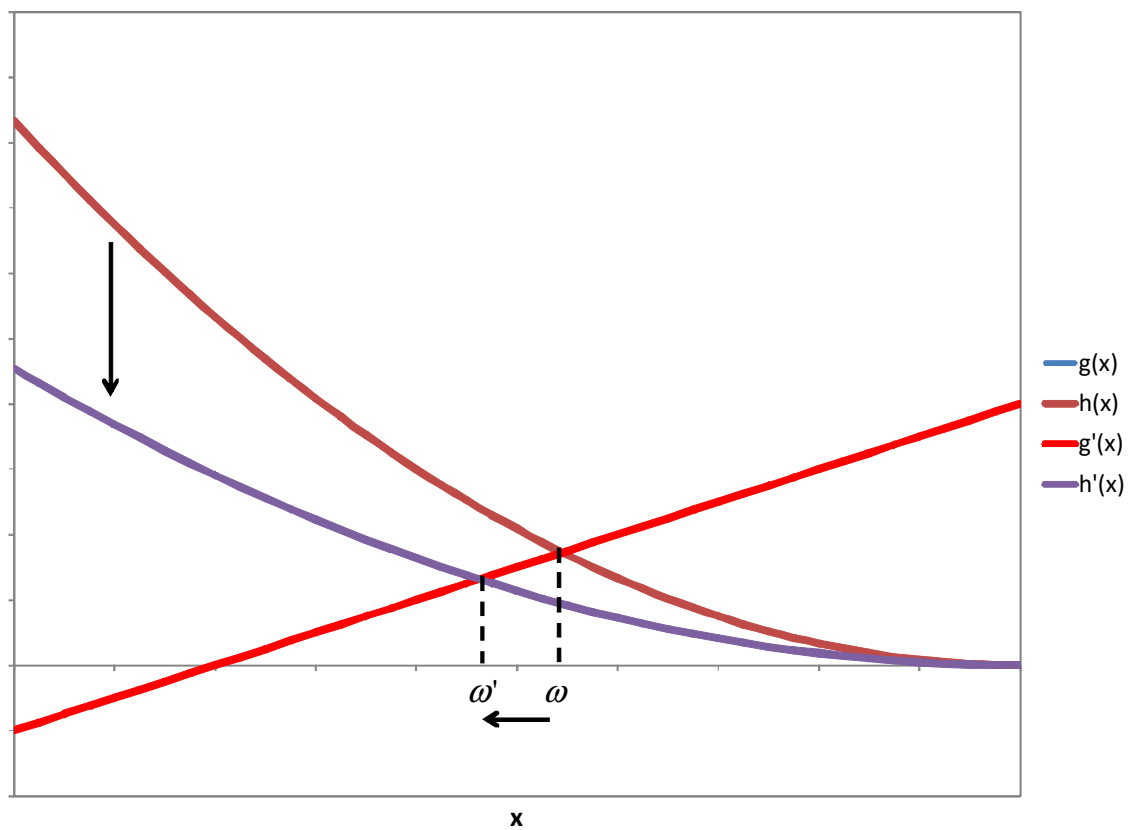
$$\uparrow \lambda \Rightarrow \uparrow \omega \Rightarrow \uparrow F(\omega) \Rightarrow \downarrow 1 - F(\omega) \Rightarrow \downarrow \uparrow \theta_u = \lambda[1 - F(\omega)] \Rightarrow \uparrow \downarrow u = \frac{\theta_e}{\theta_e + \theta_u}$$



Change in α

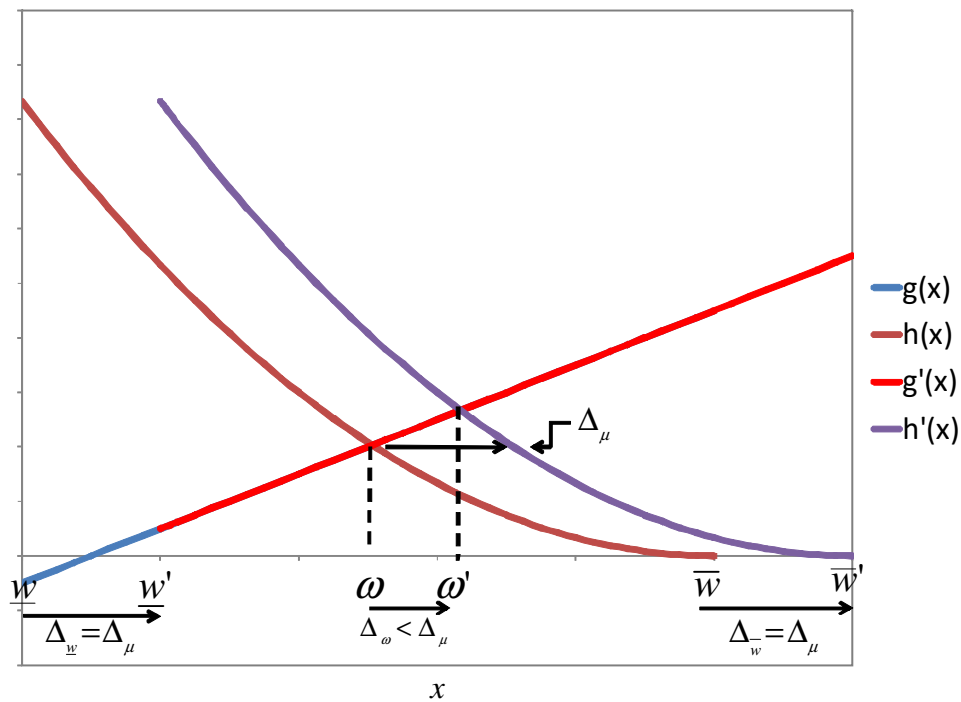
$$\frac{\partial \omega}{\partial \alpha} = -\frac{\omega - z}{\rho + \alpha + \theta_u} < 0$$

$$\left. \begin{array}{l} \uparrow \alpha \Rightarrow \downarrow \omega \Rightarrow \downarrow F(\omega) \Rightarrow \uparrow 1 - F(\omega) \Rightarrow \uparrow \theta_u = \lambda[1 - F(\omega)] \\ \uparrow \alpha \Rightarrow \uparrow \theta_e \end{array} \right\} \Rightarrow \uparrow \downarrow u = \frac{\theta_e}{\theta_e + \theta_u}$$



A shift to the right of the wage distribution

The mean $\mu \equiv \mathbb{E}W$ increases and the support shifts too.



Then: $\uparrow \mu \Rightarrow \uparrow \omega$

Then: $\begin{cases} \uparrow \mu \Rightarrow \downarrow F(\omega; \mu) \text{ for given } \omega \\ \uparrow \omega \Rightarrow \uparrow F(\omega; \mu) \text{ for given } \mu \end{cases}$

Since $\uparrow \omega < \uparrow \mu$ we know $\downarrow F(\omega; \mu) \Rightarrow \uparrow 1 - F(\omega; \mu) \Rightarrow \uparrow \theta_u = \lambda[1 - F(\omega; \mu)] \Rightarrow \downarrow u = \frac{\theta_e}{\theta_e + \theta_u}$

Appendix

Expected Duration of Employment and Unemployment Spells

Let T_u denote the (random) duration of a completed unemployment spell, and let D_u denote its cumulative distribution function: $D_u(t) \equiv \Pr\{T_u \leq t\}$. Consider the following question: Given that the unemployment spell has already lasted t units of time, what is the probability that it will end in the next interval $h > 0$? In other words, we are interested in

$$\Pr\{T_u \in [t, t+h] \mid T_u \geq t\}.$$

Using the law of conditional probabilities we can rewrite the expression above as follows:

$$\begin{aligned} \Pr\{T_u \in [t, t+h] \mid T_u \geq t\} &= \frac{\Pr\{T_u \in [t, t+h] \text{ and } T_u \geq t\}}{\Pr\{T_u \geq t\}} \\ &= \frac{\Pr\{T_u \in [t, t+h]\}}{\Pr\{T_u \geq t\}} \\ &= \frac{\Pr\{T_u \leq t+h\} - \Pr\{T_u \leq t\}}{1 - \Pr\{T_u \leq t\}} \quad 1 \\ &= \frac{D_u(t+h) - D_u(t)}{1 - D_u(t)} \end{aligned}$$

It is standard to characterize this aspect of the distribution using the *hazard function*

$$\theta_u(t) \equiv \lim_{h \rightarrow 0} \frac{\Pr\{T_u \in [t, t+h] \mid T_u \geq t\}}{h}$$

Using our previous result we can write:

$$\begin{aligned} \theta_u(t) &= \lim_{h \rightarrow 0} \frac{D_u(t+h) - D_u(t)}{[1 - D_u(t)]h} \\ &= \frac{1}{1 - D_u(t)} \lim_{h \rightarrow 0} \frac{D_u(t+h) - D_u(t)}{h} \\ &= \frac{D'_u(t)}{1 - D_u(t)} \\ &= \frac{d_u(t)}{S_u(t)} \end{aligned}$$

where $d_u(t) \equiv D'_u(t)$ is the density function and $S_u(t) \equiv 1 - D_u(t) = \Pr\{T_u > t\}$ is the so-called *survival function*.²

¹With no mass points, $\Pr\{T_u = t\} = 0$. Then: $\Pr\{T_u \leq t\} = \Pr\{T_u < t\}$.

²The terms *hazard* and *survival* come from biostatistics, where it is standard to ask for the probability that an individual dies during interval $[t, t+h]$ given that it has already survived up to t . When talking about unemployment spells, this vocabulary is a bit counterintuitive because a higher probability of survival means a higher probability of remaining unemployed.

Notice that: $\frac{d \ln S_u(\tau)}{d\tau} = \frac{1}{S_u(\tau)} S'_u(\tau) = \frac{-D'_u(\tau)}{S_u(\tau)} = -\frac{d_u(\tau)}{S_u(\tau)} = -\theta_u(\tau)$. Then:

$$\theta_u(\tau) = -\frac{d \ln S_u(\tau)}{d\tau}$$

Integrating both sides of the expression above from 0 to t we get:

$$S_u(t) = e^{-\int_0^t \theta_u(\tau) d\tau}$$

The expression above, together with $d_u(t) = \theta_u(t)S_u(t)$, gives:

$$d_u(t) = \theta_u(t)e^{-\int_0^t \theta_u(\tau) d\tau}$$

The hazard function evaluated at t gives the *rate* at which unemployment spells are terminated, given that they have already lasted until t . In our search model, the rate at which a worker escapes unemployment equals the rate at which employment offers arrive, multiplied by the probability that the offer is accepted (conditional on arrival). Since offers arrive at rate λ and the probability of acceptance is $1 - F(\omega)$, we get:

$$\theta_u = \lambda[1 - F(\omega)].$$

Notice that the reservation wage is stationary, F does not change over time, and λ is constant. Hence, the hazard rate is independent of the elapsed duration of the unemployment spell. In this case we have $\int_0^t \theta_u(\tau) d\tau = \int_0^t \theta_u d\tau = \theta_u t$. Hence, $d_u(t) = \theta_u(t)e^{-\int_0^t \theta_u(\tau) d\tau}$ becomes:

$$d_u(t) = \theta_u e^{-\theta_u t}$$

which shows that the distribution of completed unemployment spells is exponential, with parameter θ_u . The associated survival function is

$$S_u(t) = e^{-\theta_u t}.$$

Once we've found the distribution of completed unemployment spells, we can calculate their expected duration:

$$\begin{aligned} \mathbb{E}T_u &= \int_0^\infty t d_u(t) dt \\ &= \int_0^\infty t \theta_u e^{-\theta_u t} dt. \end{aligned}$$

Integrating by parts, using $u = t$ and $v = -e^{-\theta_u t}$, we obtain:

$$\mathbb{E}T_u = \frac{1}{\theta_u}.$$

It can also be shown that:

$$VarT_u = \frac{1}{\theta_u^2}.$$

We can repeat the calculations for employment spells. Letting T_e denote the duration of completed employment spells, and using analogous notation, we obtain:

$$\theta_e = \alpha$$

$$d_e(t) = \alpha e^{-\alpha t}$$

$$D_e(t) = 1 - e^{-\alpha t}$$

$$S_e(t) = e^{-\alpha t}$$

$$\mathbb{E}T_e = \frac{1}{\alpha}$$

and

$$VarT_e = \frac{1}{\alpha^2}.$$

Determination of the reservation wage: graphical representation

In class we defined:

$$g(x) = x - z \quad \forall x \in [\underline{w}, \overline{w}]$$

$$h(x) = \frac{\lambda}{\rho + \alpha} \int_x^{\overline{w}} (w - x) f(w) dw \quad \forall x \in [\underline{w}, \overline{w}]$$

More generally, we can define the functions above over all the real numbers, as follows:

$$g(x) = x - z \quad \forall x \in \mathbb{R}$$

$$h(x) = \frac{\lambda}{\rho + \alpha} \int_x^{\infty} (w - x) f(w) dw \quad \forall x \in \mathbb{R}$$

The function g is obviously a straight line with slope equal to 1 and intercept (at $x = 0$) equal to $-z$.

For h we get three different parts.

Part 1. $x < \underline{w}$

In this case, we have:

$$\begin{aligned} h(x) &= \frac{\lambda}{\rho + \alpha} \int_x^{\infty} (w - x) f(w) dw \\ h(x) &= \frac{\lambda}{\rho + \alpha} \left[\underbrace{\int_x^{\underline{w}} (w - x) f(w) dw}_{=0} + \int_{\underline{w}}^{\overline{w}} (w - x) f(w) dw + \underbrace{\int_{\overline{w}}^{\infty} (w - x) f(w) dw}_{=0} \right] \\ h(x) &= \frac{\lambda}{\rho + \alpha} \int_{\underline{w}}^{\overline{w}} (w - x) f(w) dw \\ h(x) &= \frac{\lambda}{\rho + \alpha} \left[\int_{\underline{w}}^{\overline{w}} w f(w) dw - x \int_{\underline{w}}^{\overline{w}} f(w) dw \right] \\ h(x) &= \frac{\lambda}{\rho + \alpha} (\mathbb{E}W - x) \end{aligned}$$

Then:

$$h(x) = \frac{\lambda}{\rho + \alpha} (\mu - x) \quad \forall x \in (-\infty, \underline{w})$$

The expression above is a straight line with slope equal to $-\frac{\lambda}{\rho + \alpha}$ and intercept (at $x = 0$) equal to $\frac{\lambda}{\rho + \alpha} \mu$.

Part 2. $x \in [\underline{w}, \overline{w}]$

In this case, we have:

$$\begin{aligned} h(x) &= \frac{\lambda}{\rho + \alpha} \int_x^{\infty} (w - x) f(w) dw \\ h(x) &= \frac{\lambda}{\rho + \alpha} \int_x^{\overline{w}} (w - x) f(w) dw + \underbrace{\frac{\lambda}{\rho + \alpha} \int_{\overline{w}}^{\infty} (w - x) f(w) dw}_{=0} \end{aligned}$$

Then:

$$h(x) = \frac{\lambda}{\rho + \alpha} \int_x^{\overline{w}} (w - x) f(w) dw \quad \forall x \in [\underline{w}, \overline{w}]$$

This is the part we analyzed in class

Part 3. $x > \overline{w}$

In this case, we have:

$$h(x) = \frac{\lambda}{\rho+\alpha} \int_x^\infty (w-x) \underbrace{f(w)}_{=0} dw$$

Then:

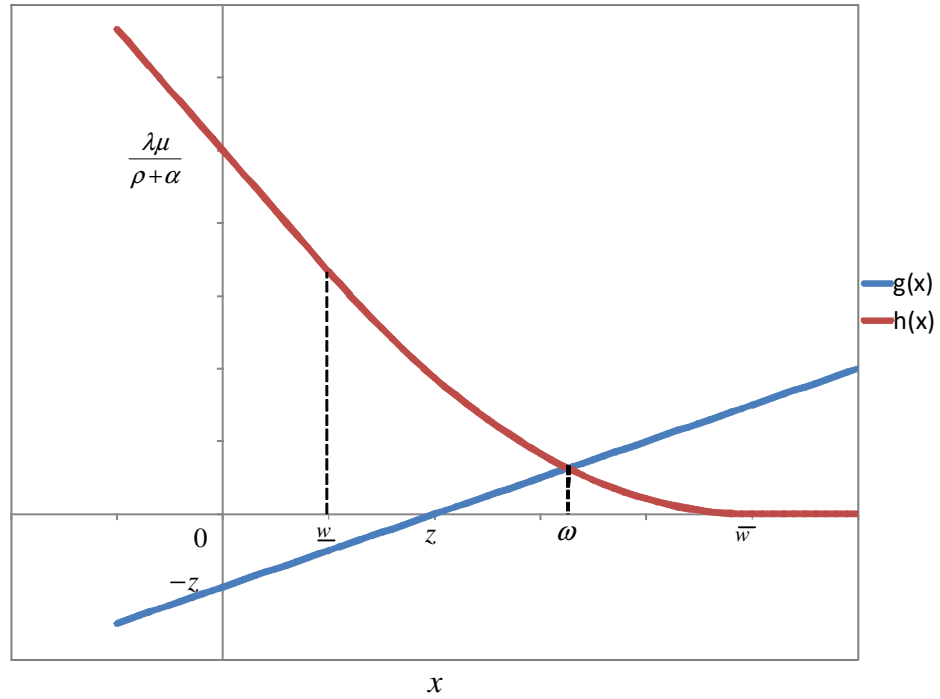
$$h(x) = 0 \quad \forall x > \overline{w}$$

Summarizing

$$g(x) = x - z \quad \forall x \in \mathbb{R}$$

$$h(x) = \begin{cases} \frac{\lambda}{\rho+\alpha} (\mu - x) & \forall x < \underline{w} \\ \frac{\lambda}{\rho+\alpha} \int_x^{\overline{w}} (w-x) f(w) dw & \forall x \in [\underline{w}, \overline{w}] \\ 0 & \forall x > \overline{w} \end{cases}$$

Plot



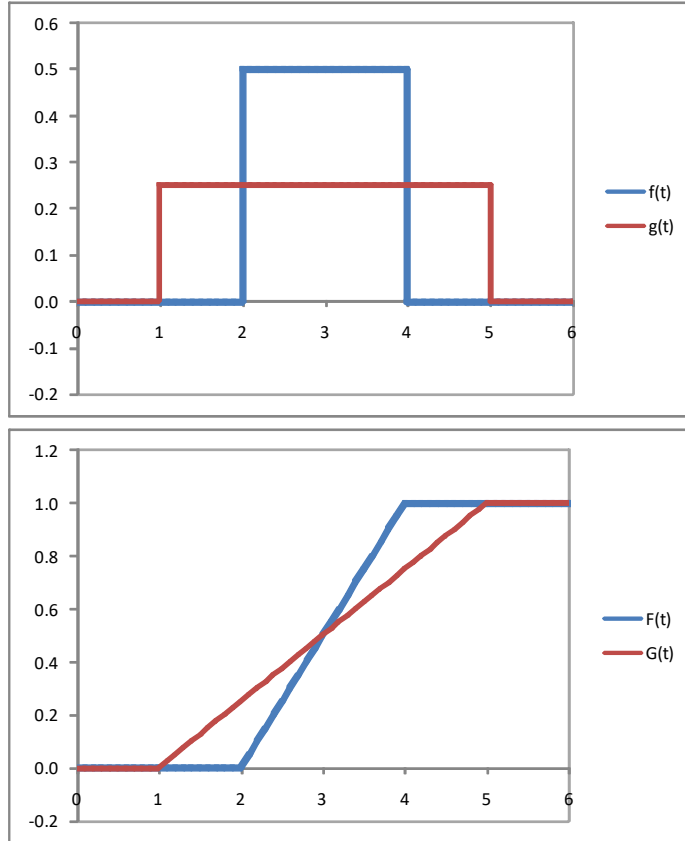
Mean-Preserving Spreads and the Reservation Wage

In what follows, we are going to consider non-negative random variables with cumulative distribution functions (cdf's) of the form $F : \mathbb{R}_+ \rightarrow [0, 1]$, with positive density $f(x) \equiv F'(x) > 0$ on a closed interval $[\underline{x}, \bar{x}]$, where $\bar{x} > \underline{x} \geq 0$, and $f(x) = 0 \ \forall x \notin [\underline{x}, \bar{x}]$. Notice that $F(x) = 0 \ \forall x \leq \underline{x}$ and $F(x) = 1 \ \forall x \geq \bar{x}$.

Definition (Informal): Simple Mean-Preserving Spread (MPS)

A distribution G is obtained by a simple mean-preserving spread of F if it is derived from F by shifting part of the mass of the distribution toward the tails, keeping the mean constant.

The figure below provides an example where a uniform distribution (G) is obtained from another uniform distribution (F) through a simple MPS. We plot the probability density functions (pdf's) and the cdf's for both random variables.



Theorem: Rothschild and Stiglitz (1970).

Consider two random variables X and Y with distributions F and G , respectively. Assume that X and Y have the same mean. Then, the following statements are equivalent:

- (a) G can be obtained from F by a sequence of simple MPS.
- (b) $\int_0^x G(t)dt \geq \int_0^x F(t)dt \quad \forall x \in \mathbb{R}_+$.
- (c) There is a random variable Z with $\mathbb{E}\{Z|X\} = 0$ such that $Y \stackrel{d}{=} X + Z$.
- (d) $\int_0^\infty u(x)f(x)dx \geq \int_0^\infty u(x)g(x)dx$ for any bounded concave fn. $u : \mathbb{R}_+ \rightarrow \mathbb{R}$.

Proof. See:

Rothschild, M. and Stiglitz, J. (1970). "Increasing Risk I: A Definition", *Journal of Economic Theory*, Vol. 2, No. 3 (September), pp. 225-243.

Rothschild, M. and Stiglitz, J. (1971). "Increasing Risk II: Its Economic Consequences", *Journal of Economic Theory*, Vol. 3, No. 1 (March), pp. 66-84.

Rothschild, M. and Stiglitz, J. (1972). "Addendum to 'Increasing Risk: I. A definition'", *Journal of Economic Theory*, Vol. 5, No. 2 (October), p. 206.

Rasmusen, E. and Petrakis, E. (1992). "Defining the Mean-Preserving Spread: 3-pt versus 4-pt," in *Decision Making Under Risk and Uncertainty: New Models and Empirical Findings*, John Geweke (editor), Amsterdam: Kluwer.

Machina, M. and Pratt, J. (1997). "Increasing Risk, Some Direct Constructions," *Journal of Risk and Uncertainty*, Vol. 14, pp. 103-127.

Leshno, M., Levy, H. and Spector, Y. (1997). "A Comment on Rothschild and Stiglitz's 'Increasing Risk: I. A Definition'," *Journal of Economic Theory*, Vol. 77, No. 1 (November), pp. 223-228.

Müller, Alfred, (1998). "Comparing Risks with Unbounded Distributions," *Journal of Mathematical Economics*, Vol. 30, No. 2 (September), pp. 229-239.

■

Remark 1. The symbol $\stackrel{d}{=}$ indicates equality *in distribution*. That is, Y is a random variable that takes the same values with the same probabilities as the random variable $X + Z$. Equality in distribution is a much weaker condition than equality: two random variables are equal if they take on the same value in every state, which is sufficient but not necessary for equality in distribution.

For example $X = \begin{cases} 0 & \text{if } s = 1 \\ 1 & \text{if } s = 2 \end{cases}$ and $Y = \begin{cases} 1 & \text{if } s = 1 \\ 0 & \text{if } s = 2 \end{cases}$ are different random variables, but they are equal in distribution when $\Pr(s = 1) = \Pr(s = 2) = \frac{1}{2}$.

Remark 2. Mean independence ($\mathbb{E}\{Z|X\} = 0$) is weaker than independence but stronger than uncorrelatedness. Independence implies mean independence, and mean independence implies uncorrelatedness, but the converse statements are not true in general.

Remark 3. Condition $\int_0^\infty u(x)f(x)dx \geq \int_0^\infty u(y)g(y)dy$ means that F second-order stochastically dominates G . It shows that any risk-averse, expected-utility maximizer will prefer lottery F to lottery G . When this happens we say that F is less risky than G or, equivalently, that G is riskier than F .

Remark 4. The cdf's F and G may cross more than once.

The part of the previous theorem that is useful for us is (b). To see the intuition for it we need first a couple of results.

Claim 1. Suppose that X is a random variable with the properties described at the beginning of this note. Let b denote a real number such that $b \geq \bar{x}$. Then, $\mathbb{E}\{X\} = b - \int_0^b F(x)dx = \int_0^b [1 - F(x)]dx$.

Proof.

We know

$$\mathbb{E}\{X\} = \int_{\underline{x}}^{\bar{x}} xf(x)dx$$

The expression above can be rewritten as follows:

$$\mathbb{E}\{X\} = \underbrace{\int_0^{\underline{x}} xf(x)dx}_{=0} + \int_{\underline{x}}^{\bar{x}} xf(x)dx + \underbrace{\int_{\bar{x}}^b xf(x)dx}_{=0}$$

Then:

$$\mathbb{E}\{X\} = \int_0^b xf(x)dx$$

Integrating by parts we get (with $u = x$ and $dv = f(x)dx$):

$$\mathbb{E}\{X\} = xF(x)\Big|_0^b - \int_0^b F(x)dx$$

$$\mathbb{E}\{X\} = [bF(b) - 0F(0)] - \int_0^b F(x)dx$$

Then:

$$\mathbb{E}\{X\} = b - \int_0^b F(x)dx$$

where we have used $F(b) = 1$, since $b \geq \bar{x}$.

The expression above can be rewritten as follows:

$$\mathbb{E}\{X\} = \int_0^b dx - \int_0^b F(x)dx$$

Then:

$$\mathbb{E}\{X\} = \int_0^b [1 - F(x)]dx$$

■

Claim 2. Consider two non-negative random variables, X and Y , with supports $[\underline{x}, \bar{x}]$ and $[\underline{y}, \bar{y}]$, and let $b > \bar{x}$ and $b > \bar{y}$. Let F and G denote the cdfs of X and Y , respectively. Assume both random variables have the same mean: $\mathbb{E}\{X\} = \mathbb{E}\{Y\}$. Then: $\int_0^b F(x)dx = \int_0^b G(x)dx$. Moreover, $\int_0^\infty F(x)dx = \int_0^\infty G(x)dx$.
Proof.

From the proof of our previous claim we know:

$$\mathbb{E}\{X\} = b - \int_0^b F(x)dx$$

$$\mathbb{E}\{Y\} = b - \int_0^b G(x)dx$$

Then, the equality of means implies:

$$\begin{aligned} b - \int_0^b F(x)dx &= b - \int_0^b G(x)dx \\ - \int_0^b F(x)dx &= - \int_0^b G(x)dx \Rightarrow \end{aligned}$$

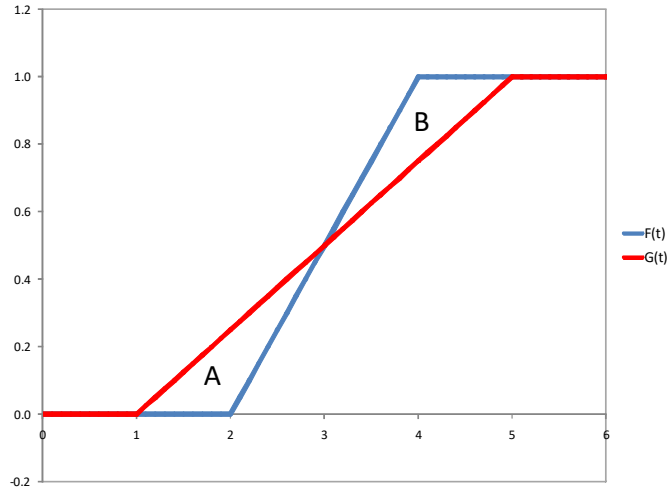
$$\int_0^b F(x)dx = \int_0^b G(x)dx$$

Since $F(x) = G(x) = 1 \ \forall x > b$, the expression above implies:

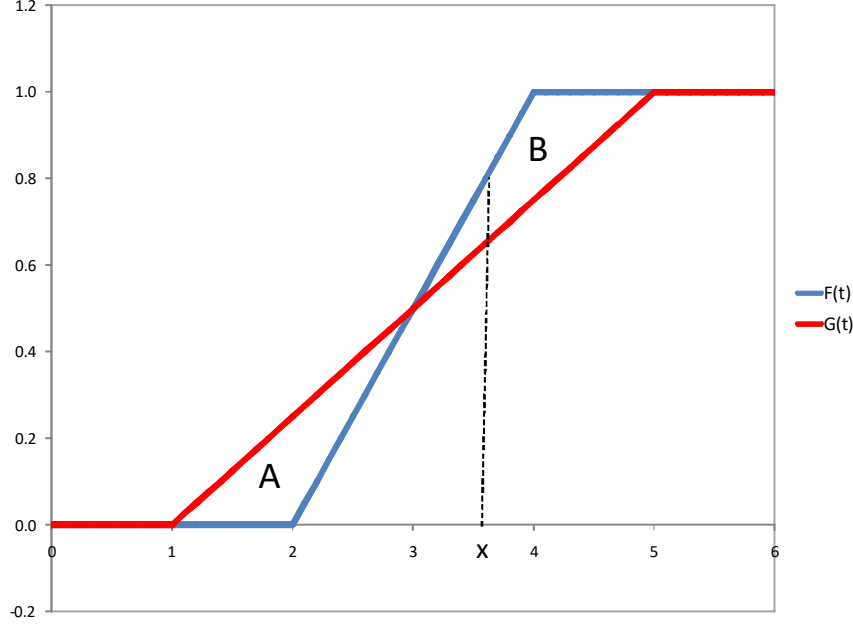
$$\int_0^\infty F(x)dx = \int_0^\infty G(x)dx$$

■

Claim 2 shows that the areas denoted by A and B in the following figure must be equal.



But then the next figure shows that the area below G from 0 to x must be bigger than the area below F . And this is the result established in part (b) of the theorem above: $\int_0^x G(t)dt \geq \int_0^x F(t)dt$.



Part (b) of Rothschild and Stiglitz's theorem can be used to show the effect of a mean-preserving spread of the wage distribution on the reservation wage. Recall that the reservation wage, ω , is implicitly defined by the following expression:

$$\omega - z = \frac{\lambda}{\rho + \alpha} \int_{\omega}^{\bar{w}} (w - \omega) f(w) dw$$

It will be convenient to rewrite the expression above in a different way. Adding and subtracting $\frac{\lambda}{\rho + \alpha} \int_{\underline{w}}^{\omega} (w - \omega) f(w) dw$ from the right-hand side we get:

$$\begin{aligned} \omega - z &= \frac{\lambda}{\rho + \alpha} \int_{\omega}^{\bar{w}} (w - \omega) f(w) dw + \frac{\lambda}{\rho + \alpha} \int_{\underline{w}}^{\omega} (w - \omega) f(w) dw - \frac{\lambda}{\rho + \alpha} \int_{\underline{w}}^{\omega} (w - \omega) f(w) dw \\ \omega - z &= \frac{\lambda}{\rho + \alpha} \int_{\underline{w}}^{\bar{w}} (w - \omega) f(w) dw - \frac{\lambda}{\rho + \alpha} \int_{\underline{w}}^{\omega} (w - \omega) f(w) dw \\ \omega - z &= \frac{\lambda}{\rho + \alpha} \left(\int_{\underline{w}}^{\bar{w}} w f(w) dw - \omega \int_{\underline{w}}^{\bar{w}} f(w) dw \right) - \frac{\lambda}{\rho + \alpha} \int_{\underline{w}}^{\omega} (w - \omega) f(w) dw \\ \omega - z &= \frac{\lambda}{\rho + \alpha} (\mathbb{E}W - \omega) - \frac{\lambda}{\rho + \alpha} \int_{\underline{w}}^{\omega} (w - \omega) f(w) dw \\ \omega - z &= \frac{\lambda}{\rho + \alpha} (\mu - \omega + z - z) - \frac{\lambda}{\rho + \alpha} \int_{\underline{w}}^{\omega} (w - \omega) f(w) dw \\ (\omega - z) \left(1 + \frac{\lambda}{\rho + \alpha} \right) &= \frac{\lambda}{\rho + \alpha} (\mu - z) - \frac{\lambda}{\rho + \alpha} \int_{\underline{w}}^{\omega} (w - \omega) f(w) dw \\ (\omega - z) \frac{\rho + \alpha + \lambda}{\rho + \alpha} &= \frac{\lambda}{\rho + \alpha} (\mu - z) - \frac{\lambda}{\rho + \alpha} \int_{\underline{w}}^{\omega} (w - \omega) f(w) dw \end{aligned}$$

Then:

$$\omega - z = \frac{\lambda}{\rho + \alpha + \lambda}(\mu - z) - \frac{\lambda}{\rho + \alpha + \lambda} \int_{\underline{w}}^{\omega} (w - \omega) f(w) dw$$

Integrating the second term of the right-hand side by parts, with $u = w - \omega$ and $dv = f(w)dw$, we get:

$$\begin{aligned} \int_{\underline{w}}^{\omega} (w - \omega) f(w) dw &= (w - \omega) F(w) \Big|_{\underline{w}}^{\omega} - \int_{\underline{w}}^{\omega} F(w) dw \\ \int_{\underline{w}}^{\omega} (w - \omega) f(w) dw &= (\omega - \omega) F(\omega) - (\underline{w} - \omega) F(\underline{w}) - \int_{\underline{w}}^{\omega} F(w) dw \\ \int_{\underline{w}}^{\omega} (w - \omega) f(w) dw &= 0 \times F(\omega) - (\underline{w} - \omega) \times 0 - \int_{\underline{w}}^{\omega} F(w) dw \end{aligned}$$

Then:

$$\int_{\underline{w}}^{\omega} (w - \omega) f(w) dw = - \int_{\underline{w}}^{\omega} F(w) dw$$

Combining the previous two expressions we get:

$$\omega - z = \frac{\lambda}{\rho + \alpha + \lambda}(\mu - z) + \frac{\lambda}{\rho + \alpha + \lambda} \int_{\underline{w}}^{\omega} F(w) dw$$

But $\int_{\underline{w}}^{\omega} F(w) dw = \int_0^{\omega} F(w) dw$, since $F(w) = 0 \ \forall w \leq \underline{w}$. Then:

$$\omega - z = \frac{\lambda}{\rho + \alpha + \lambda}(\mu - z) + \frac{\lambda}{\rho + \alpha + \lambda} \int_0^{\omega} F(w) dw$$

Suppose now that the wage distribution changes from F to G , where G is obtained from F through a mean-preserving spread (in other words, the wage distribution becomes riskier). Then μ remains unchanged, and the term $\int_0^{\omega} F(w) dw$ in the right-hand side of the expression above changes to $\int_0^{\omega} G(w) dw$. But we know from Rothschild and Stiglitz's theorem that $\int_0^{\omega} G(w) dw \geq \int_0^{\omega} F(w) dw \ \forall \omega \geq 0$. Hence, the right-hand side of the expression above increases. But this requires an increase in the left-hand side. Hence, ω must increase. We conclude that a riskier wage distribution induces an increase in the reservation wage. The intuition for this result is that the higher incidence of very good wage offers induced by the MPS increases the value of searching for a job, while the higher incidence of very bad offers is less detrimental because these offers will be rejected anyway (the option to work will not be exercised at such low wages).